Argand Diagrams

Solutions

Chapter Overview

Complex numbers can be represented as points in a plane.

1. Plot a number using its real and imaginary parts.
2. Describe it using a distance and an angle.
3. Multiply and divide using modulus–argument form.
4. Interpret equations and inequalities as loci and regions.

Angles are in radians unless stated otherwise.

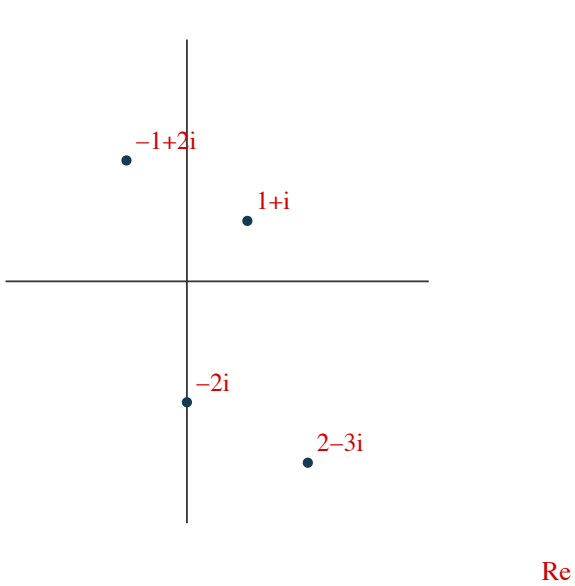
Argand Diagrams

Solution 3

Plot $1+i,\ 2-3i,\ -1+2i,\ -2i$.

For $z = x + iy$, the horizontal coordinate is $\operatorname{Re} z = x$, and the vertical coordinate is $\operatorname{Im} z = y$.

Sketch



But why visualise complex numbers?

Solution 4

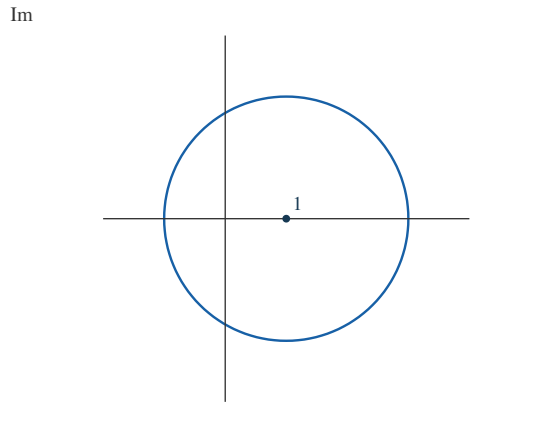
Geometry behind algebra

For a polynomial with real coefficients, non-real roots occur in conjugate pairs. On an Argand diagram they reflect in the real axis.

The four fourth roots of $1+i$ lie on a circle and are separated by angles of $\pi/2$.

The equation $|z-1| = 2$ describes a circle with centre $(1, 0)$ and radius 2.

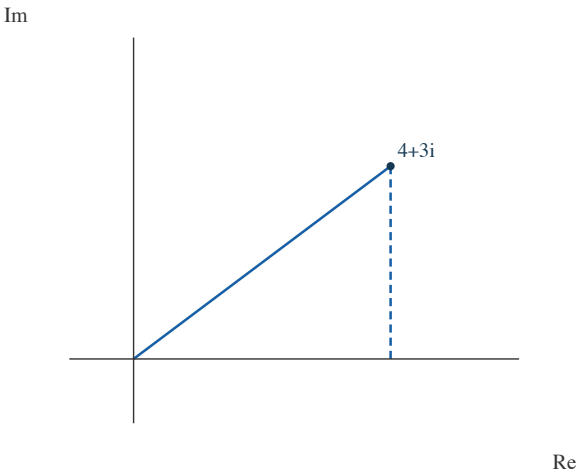
Complex-number iteration also generates geometric patterns such as the Mandelbrot set.



Modulus and Argument

Solution 5

For $z = 4 + 3i$, find its modulus r and argument θ.



Modulus and argument

$r = |z| = \sqrt{4^2 + 3^2} = 5$

$\theta = \tan^{-1}(3/4) \approx 0.644$

The principal argument satisfies $-\pi < \arg z \leq \pi$. The argument of zero is undefined.

In general, $|x + iy| = \sqrt{x^2 + y^2}$. Use the quadrant to choose the correct argument.

Examples

Solution 6

1. Find the modulus and principal argument of

   $5 + 12i$

1

$|z| = 13, \quad \arg z \approx 1.18$

2. Find the modulus and principal argument of

   $-1 + i$

2

$|z| = \sqrt{2}, \quad \arg z = 3\pi/4$

3. Find the modulus and principal argument of

   $-2i$

3

$|z| = 2, \quad \arg z = -\pi/2$

4. Find the modulus and principal argument of

   $-1 - 3i$

4

$|z| = \sqrt{10}, \quad \arg z = -\pi + \tan^{-1} 3$

Test Your Understanding — Solution 7

Given $z = 2 - 3i$:

a. Find z^2.

b. Find $|z^2|$.

c. Find $\arg(z^2)$.

d. Plot z and z^2.

a

$z^2 = (2 - 3i)^2 = -5 - 12i$

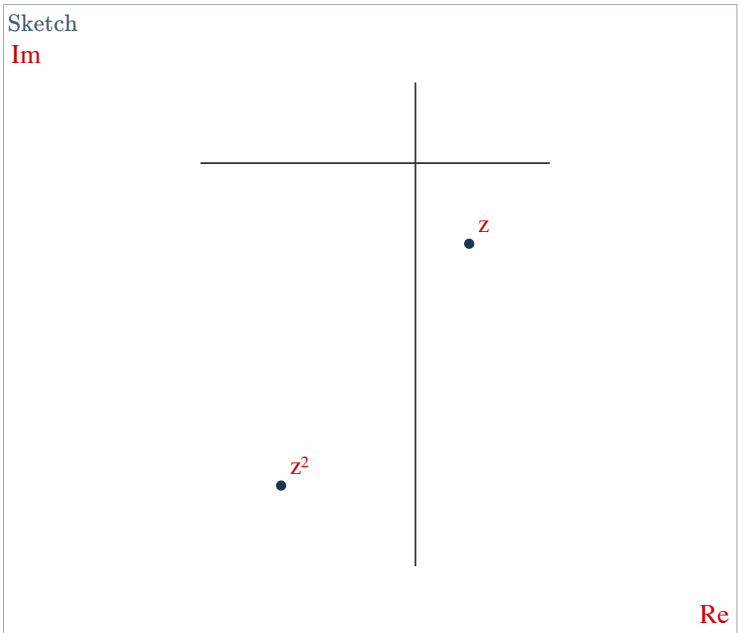
b

$|z^2| = \sqrt{25 + 144} = 13$

c

$\arg(z^2) = -\pi + \tan^{-1}(12/5) \approx -1.97$

Sketch



Exercise 2B Pearson Core Pure 1, p. 21-23 — Solution 8

3. Plot $z = -40 - 9i$ and find its principal argument.

4. Given $z = 3 + 4i$, find z^2, its modulus and argument, and plot both.

3 $\arg z = -\pi + \tan^{-1}(9/40) \approx -2.92$

Sketch

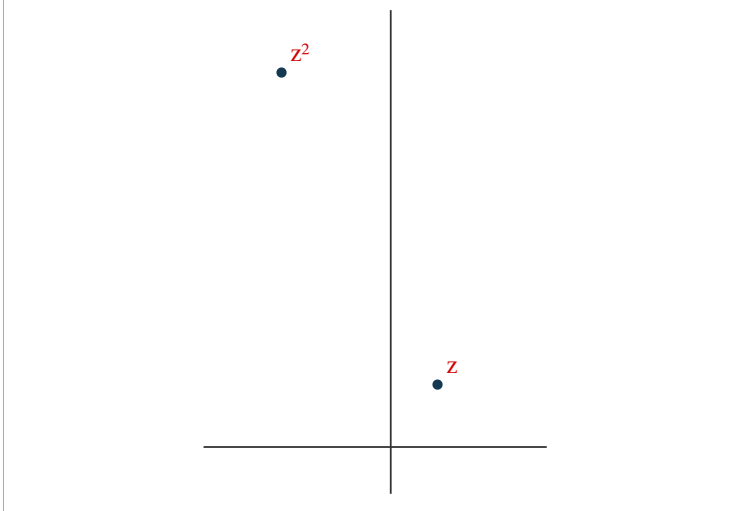


4

$z^2 = -7 + 24i, \quad |z^2| = 25$

$\arg(z^2) = \pi - \tan^{-1}(24/7) \approx 1.85$

Sketch



Exercise 2B Pearson Core Pure 1, p. 21-23 — Solution 9

5. Let $z_1 = 4 + 6i$, $z_2 = 1 + i$. Find z_1/z_2, its modulus and argument.

7. Let $z = 26/(2 - 3i)$. Find z, z^2, $|z|$ and $\arg(z^2)$.

5

$$\frac{4+6i}{1+i} = 5 + i$$

$$|z_1/z_2| = \sqrt{26}$$

$$\arg(z_1/z_2) = \tan^{-1}(1/5) \approx 0.20$$

7

$$z = 4 + 6i, \quad z^2 = -20 + 48i$$

$$|z| = 2\sqrt{13}$$

$$\arg(z^2) = \pi - \tan^{-1}(12/5) \approx 1.97$$

Exercise 2B Pearson Core Pure 1, p. 21-23 — Solution 10

8. Let $z_1 = 4 + 2i$, $z_2 = 2 + 4i$, $z_3 = a + bi$.

a. Find $|z_1 + z_2|$.

b. Express $w = z_1 z_3/z_2$ in terms of a, b.

c. Given $w = 21/5 - 22i/5$, find a, b.

d. Find $\arg w$.

a

$$|6 + 6i| = 6\sqrt{2}$$

b

$$\frac{z_1}{z_2} = \frac{4}{5} - \frac{3}{5}i$$

$$w = \frac{4a+3b}{5} + i\frac{-3a+4b}{5}$$

c

$$4a + 3b = 21, \quad -3a + 4b = -22$$

$$a = 6, \quad b = -1$$

d

$$\arg w = -\tan^{-1}(22/21) \approx -0.81$$

9. Let $w = 6 + 3i$. Find its modulus and argument. If $\arg(\lambda + 5i + w) = \pi/4$, find real λ.

9

$|w| = 3\sqrt{5}, \quad \arg w = \tan^{-1}(1/2)$

$\lambda + 5i + w = (\lambda + 6) + 8i$

$\lambda + 6 = 8 \Rightarrow \lambda = 2$

11. Let $w = k + i$, $z = -4 + 5ki$. Given $\arg(w + z) = 2\pi/3$, find real k.

11

$$\frac{1 + 5k}{k - 4} = -\sqrt{3}$$

$$k = \frac{4\sqrt{3} - 1}{5 + \sqrt{3}}$$

$$= \frac{21\sqrt{3} - 17}{22}$$

This gives negative real part and positive imaginary part, as required.

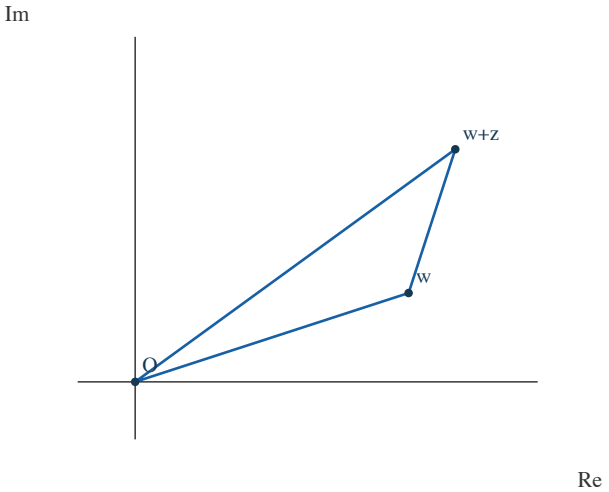
12. Given $|w| = 5$, $\arg w = \pi/10$, $\arg z = 2\pi/5$, and $\arg(w + z) = \pi/5$, find $|z|$.

Modulus

Resolve perpendicular to the direction of $w + z$:

$|z| \sin(\pi/5) = 5 \sin(\pi/10)$

$$|z| = \frac{5 \sin(\pi/10)}{\sin(\pi/5)} \approx 2.63$$



Modulus-Argument Form

Solution 13

If $z = x + iy$, with modulus r and argument θ, then

$x = r\cos\theta, \qquad y = r\sin\theta$

Modulus–argument form

$z = r(\cos\theta + i\sin\theta)$

Use the full quadrant information when finding θ.

Angles differing by 2π represent the same direction.

The principal argument is chosen from $(-\pi, \pi]$.

Example

Solution 14

1. Express $-\sqrt{3} + i$ in modulus–argument form.

1

$r = 2, \quad \theta = 5\pi/6$

$z = 2(\cos(5\pi/6) + i\sin(5\pi/6))$

2. Express $-1 - \sqrt{3}i$ in modulus–argument form.

2

$r = 2, \quad \theta = -2\pi/3$

$z = 2(\cos(-2\pi/3) + i\sin(-2\pi/3))$

Initial questions

Solution 15

1. Convert to modulus–argument form.
   a. $2 + 2i$
   b. $3i$
   c. $-3 + 4i$
   d. $1 - \sqrt{3}i$
   e. $-2 - 5i$
   f. -20
   g. $7 - 24i$
   h. $-5 + 5i$

2. Convert to modulus–argument form.
   a. $3/(1 + i\sqrt{3})$
   b. $1/(2 - i)$
   c. $(1 + i)/(1 - i)$

3. Convert to Cartesian form, where $\operatorname{cis}\theta = \cos\theta + i\sin\theta$.
   a. $5\operatorname{cis}(\pi/2)$
   b. $\frac{1}{2}\operatorname{cis}(\pi/6)$
   c. $6\operatorname{cis}(5\pi/6)$
   d. $3\operatorname{cis}(-2\pi/3)$
   e. $2\sqrt{2}\operatorname{cis}(-\pi/4)$
   f. $-4\operatorname{cis}(7\pi/6)$

Solutions

Solution 16

Exercise answers: 1

Each pair is (r, θ).

a. $(2\sqrt{2}, \pi/4)$
b. $(3, \pi/2)$
c. $(5, \pi - \tan^{-1}(4/3))$
d. $(2, -\pi/3)$
e. $(\sqrt{29}, -\pi + \tan^{-1}(5/2))$
f. $(20, \pi)$
g. $(25, -\tan^{-1}(24/7))$
h. $(5\sqrt{2}, 3\pi/4)$

Exercise answers: 2

a. $(3/2, -\pi/3)$
b. $(1/\sqrt{5}, \tan^{-1}(1/2))$
c. $(1, \pi/2)$

Exercise answers: 3

a. $5i$
b. $\sqrt{3}/4 + i/4$
c. $-3\sqrt{3} + 3i$
d. $-3/2 - 3\sqrt{3}i/2$
e. $2 - 2i$
f. $2\sqrt{3} + 2i$

Exercise 2C Pearson Core Pure 1, p. 24 — Solution 17

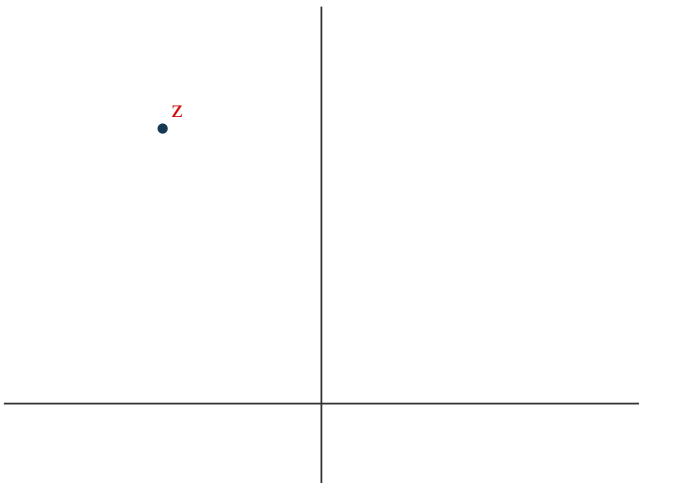
4. Write $4\,\mathrm{cis}(2\pi/3)$ in Cartesian form and plot it.

4

$-2 + 2\sqrt{3}i$

Sketch

Im



5. Find the number with modulus 7 and argument $11\pi/6$.

5

$7\sqrt{3}/2 - 7i/2$

6. Find the number with modulus 5 and argument $-4\pi/3$.

6

$-5/2 + 5\sqrt{3}i/2$

Multiplying and Dividing Complex Numbers — Solution 18

For $z_1 = r_1\,\mathrm{cis}\,\theta_1$, $z_2 = r_2\,\mathrm{cis}\,\theta_2$:

Product

$z_1 z_2 = r_1 r_2\,\mathrm{cis}(\theta_1 + \theta_2)$

Moduli multiply; arguments add.

Quotient

$\dfrac{z_1}{z_2} = \dfrac{r_1}{r_2}\,\mathrm{cis}(\theta_1 - \theta_2)$

Moduli divide; arguments subtract. Here $z_2 \neq 0$.

The addition formulae for sine and cosine give

$(\cos\alpha + i\sin\alpha)(\cos\beta + i\sin\beta)$

$= \cos(\alpha+\beta) + i\sin(\alpha+\beta)$

Reduce the resulting argument modulo 2π when a principal argument is requested.

Example

$(1+i)(1+\sqrt{3}i) = 2\sqrt{2}\,\mathrm{cis}(7\pi/12)$

Multiplying and Dividing Complex Numbers

Solution 19

Simplify in Cartesian form:

$$\frac{\sqrt{2}\,\mathrm{cis}(\pi/12)}{2\,\mathrm{cis}(5\pi/6)}$$

Answer

$$= \frac{1}{\sqrt{2}}\,\mathrm{cis}(-3\pi/4)$$

$$= -\frac{1}{2} - \frac{1}{2}i$$

Exercise 2D Pearson Core Pure 1, p. 27-28

Solution 20

4a. Simplify $\mathrm{cis}(5\theta)/\,\mathrm{cis}(2\theta)$.

a

$\cos 3\theta + i \sin 3\theta$

4b. Simplify $\sqrt{2}\,\mathrm{cis}(\pi/2)/(\frac{1}{2}\,\mathrm{cis}(\pi/4))$.

b

$2\sqrt{2}\,\mathrm{cis}(\pi/4) = 2 + 2i$

4c. Simplify $3\,\mathrm{cis}(\pi/3)/(4\,\mathrm{cis}(5\pi/6))$.

c

$\frac{3}{4}\,\mathrm{cis}(-\pi/2) = -3i/4$

4d. Simplify $(\cos 2\theta - i \sin 2\theta)/\,\mathrm{cis}(3\theta)$.

d

$\cos 5\theta - i \sin 5\theta$

Exercise 2D Pearson Core Pure 1, p. 27-28 — Solution 21

5. Let $z = -9 + 3\sqrt{3}i$ and $w = \sqrt{3}\,\mathrm{cis}(7\pi/12)$.
   a. Express z in modulus–argument form.
   b. Find zw and z/w in that form.

a

$z = 6\sqrt{3}\,\mathrm{cis}(5\pi/6)$

b

$zw = 18\,\mathrm{cis}(-7\pi/12)$

$z/w = 6\,\mathrm{cis}(\pi/4)$

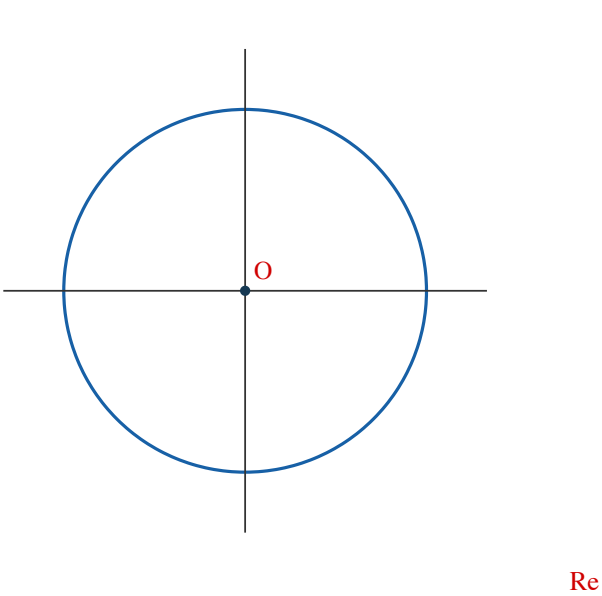
Loci — Solution 22

What is the locus $|z| = 3$?

Locus

All points at distance 3 from the origin:
a circle with centre $(0, 0)$, radius 3.

Sketch



A quick reminder...

Solution 23

Let $z = x + iy$.

Distance from origin

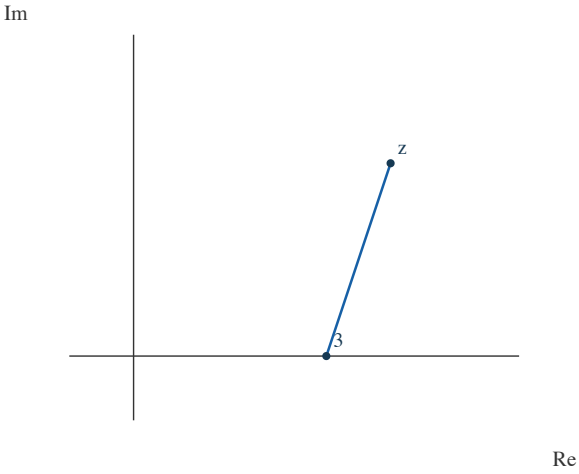
$|z| = \sqrt{x^2 + y^2}$

Distance from (3,0)

$|z - 3| = \sqrt{(x-3)^2 + y^2}$

In general, $|z - a - bi|$ is the distance from (a, b).

Therefore $|z - a - bi| = r$, with $r > 0$, is a circle.



Circular loci

Solution 24

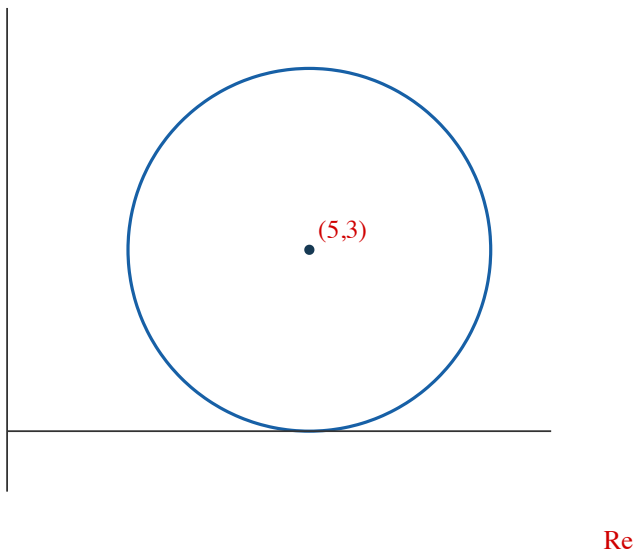
Sketch $|z - 5 - 3i| = 3$ and give its Cartesian equation.

Circle

Centre $(5, 3)$, radius 3.

$(x-5)^2 + (y-3)^2 = 9$

Sketch



Perpendicular bisector loci

Solution 25

Sketch $|z| = |z - 6i|$.

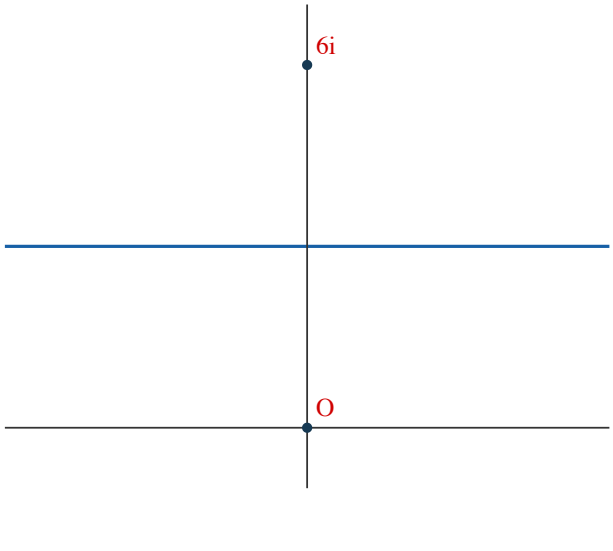
Locus

The perpendicular bisector of the segment joining $(0, 0)$ and $(0, 6)$.

$x^2 + y^2 = x^2 + (y - 6)^2$

$y = 3$

Sketch



Test Your Understanding So Far

Solution 26

Find the locus $|z - 3| = |z + i|$. Where does it meet $\text{Re}\, z = 0$?

Equation

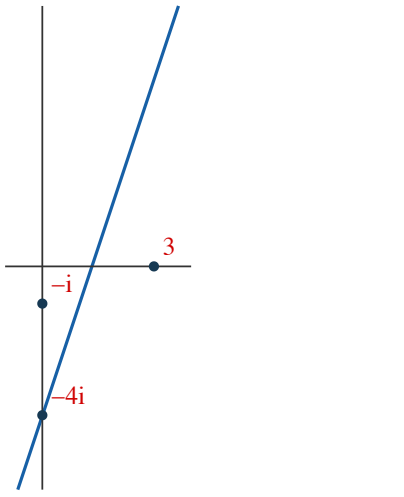
$(x - 3)^2 + y^2 = x^2 + (y + 1)^2$

$y = 3x - 4$

Intersection

$x = 0 \Rightarrow y = -4, \quad z = -4i$

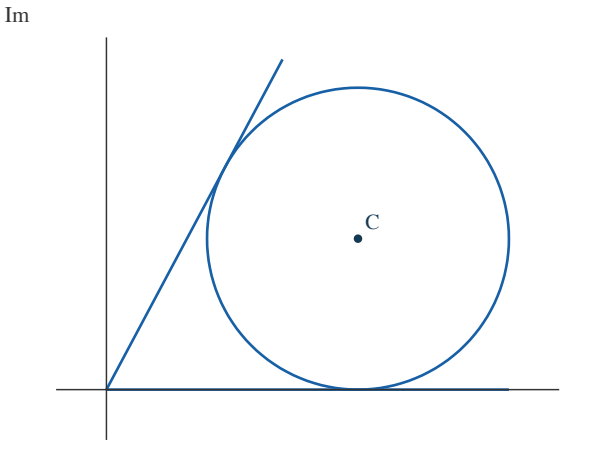
Sketch



Greatest argument and extreme moduli

Solution 27

For $|z-5-3i|=3$, find the greatest argument and the least and greatest modulus.



Argument

$\max \arg z = 2\tan^{-1}(3/5) \approx 1.081$

Modulus

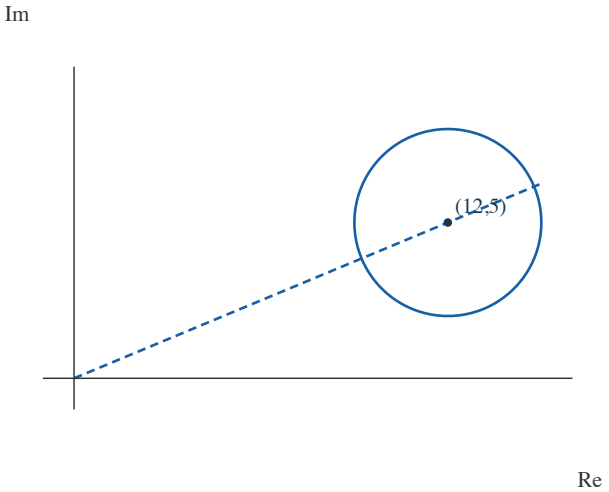
$\min |z| = \sqrt{34} - 3$

$\max |z| = \sqrt{34} + 3$

Quickfire Test Your Understanding

Solution 28

Find the least and greatest values of $|z|$ on $|z-12-5i|=3$.



Distances

$OC = \sqrt{12^2+5^2} = 13$

$\min |z| = 10, \qquad \max |z| = 16$

Least modulus on a line

Solution 29

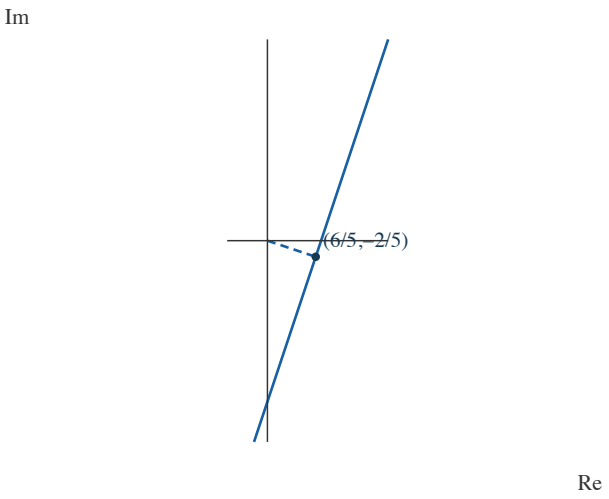
Find the least modulus on the line $|z-3| = |z+i|$.

Minimum

The line is $3x - y - 4 = 0$. The closest point is the perpendicular foot from the origin.

$$\min |z| = \frac{4}{\sqrt{10}} = \frac{2\sqrt{10}}{5}$$

$$z = \frac{6}{5} - \frac{2}{5}i$$



Rays with a fixed argument

Solution 30

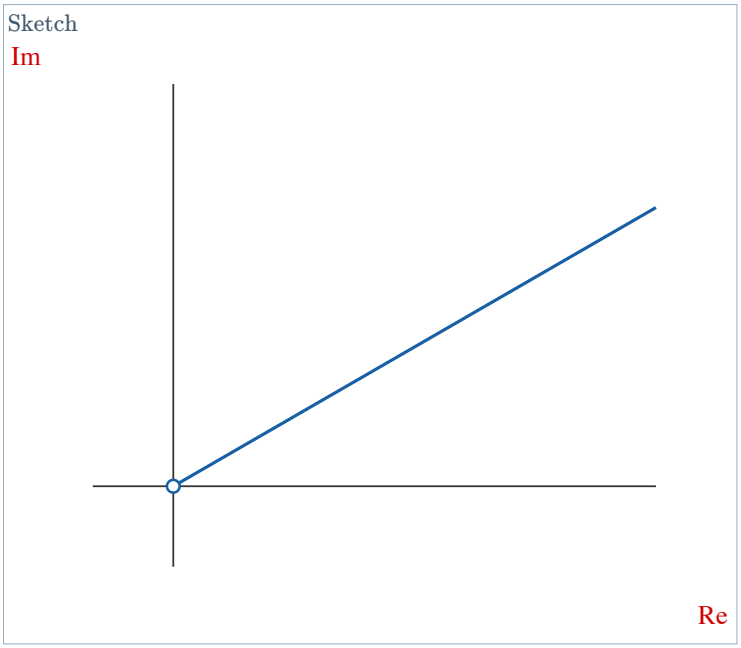
1. Sketch $\arg z = \pi/6$.
2. Sketch $\arg(z + 3 + 2i) = 3\pi/4$.

1

A ray from the origin at angle $\pi/6$, excluding the origin.

$y = x/\sqrt{3}, \quad x > 0$

Sketch

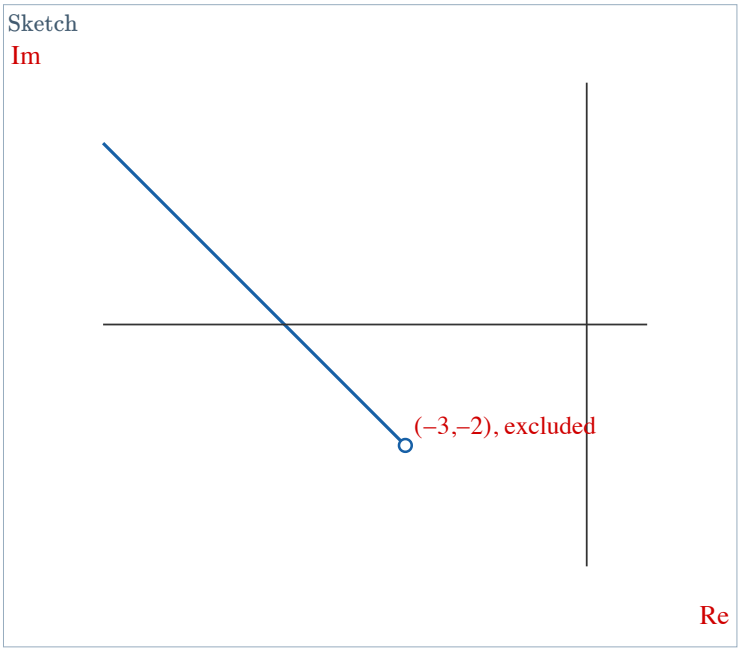


2

A ray from $(-3, -2)$, excluding its endpoint.

$y = -x - 5, \quad x < -3$

Sketch



4. For $|z-4-3i|=8$:
   a. Give its Cartesian equation.
   b. Sketch the locus.
   c. Find the greatest and least modulus.

a

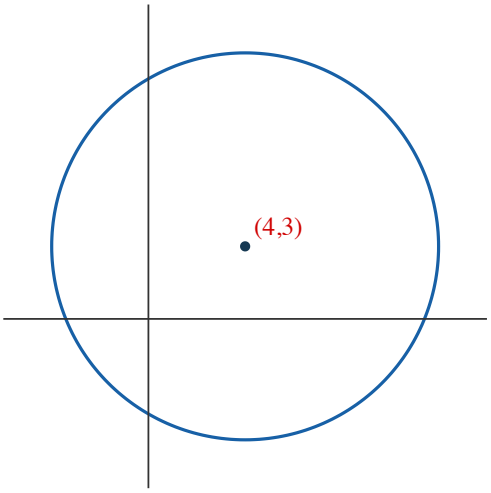
$(x-4)^2+(y-3)^2=64$

c

The origin is inside the circle, and $OC=5$.

$\max|z|=13, \quad \min|z|=3$

Sketch



5. Sketch $|z+2-2\sqrt{3}i|=2$. Find the smallest and largest principal argument.

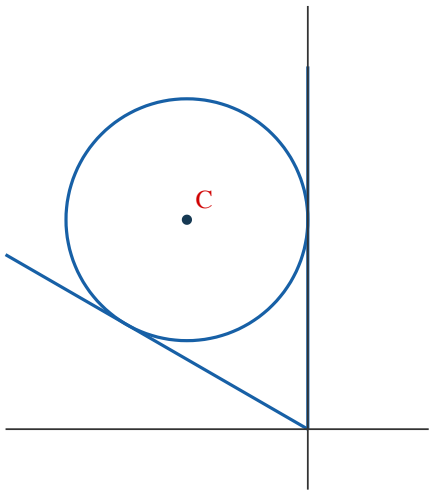
Arguments

Centre $(-2, 2\sqrt{3})$, radius 2.

The centre has argument $2\pi/3$, and tangent angle $\sin^{-1}(2/4)=\pi/6$.

$\min \arg z=\pi/2, \quad \max \arg z=5\pi/6$

Sketch



Exercise 2E Pearson Core Pure 1, p. 34-36 — Solution 33

7. Sketch $|z-3| = |z-6i|$, and find the least modulus.

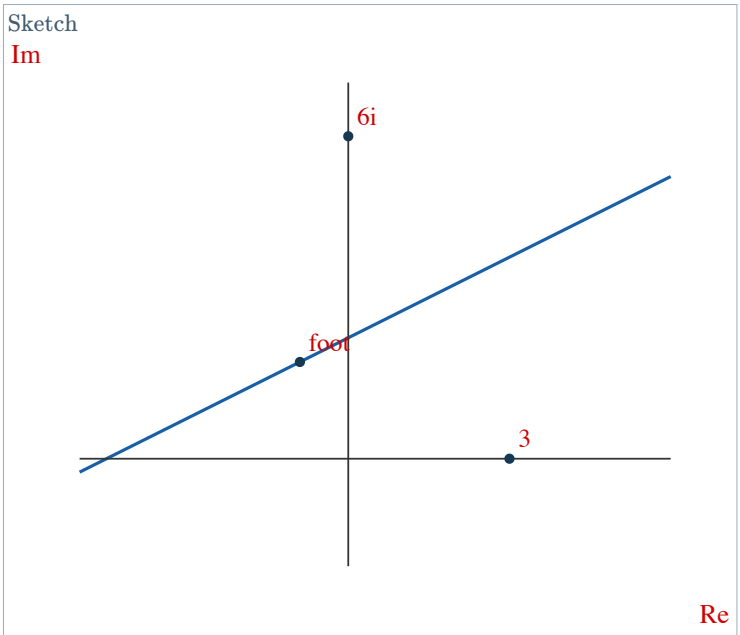
Locus and distance

$(x-3)^2 + y^2 = x^2 + (y-6)^2$

$y = \frac{1}{2}x + \frac{9}{4}$

$\min |z| = \frac{9\sqrt{5}}{10}$

Sketch



Exercise 2E Pearson Core Pure 1, p. 34-36 — Solution 34

8. Sketch $|z+3+3i| = |z-9-5i|$, and find the least modulus.

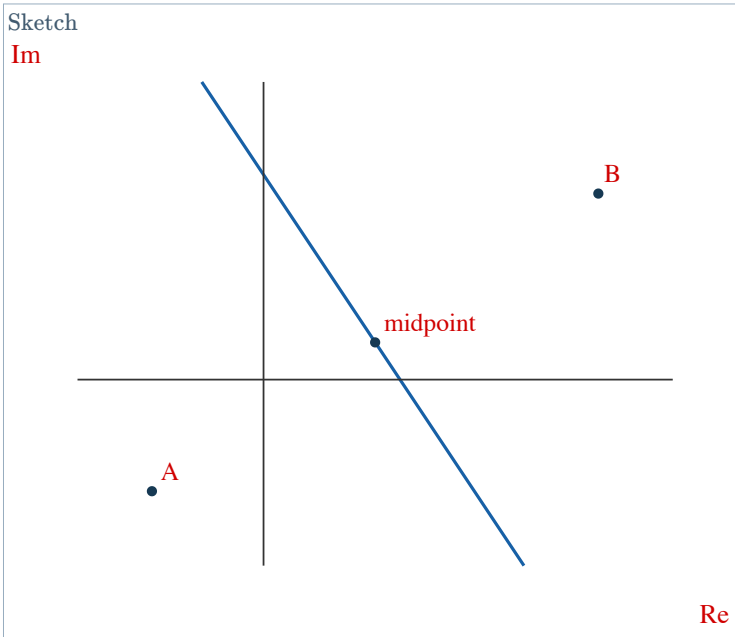
Locus and distance

The midpoint is $(3, 1)$, and the perpendicular gradient is $-3/2$.

$y = -\frac{3}{2}x + \frac{11}{2}$

$\min |z| = \frac{11}{\sqrt{13}}$

Sketch



Exercise 2E Pearson Core Pure 1, p. 34-36 — Solution 35

12. For $|z+6+6i|=4$:

a. Find the least and greatest modulus.

b. Find $-\pi<\theta<\pi$ for which the ray $\arg(z-4+2i)=\theta$ has no point on the circle.

a

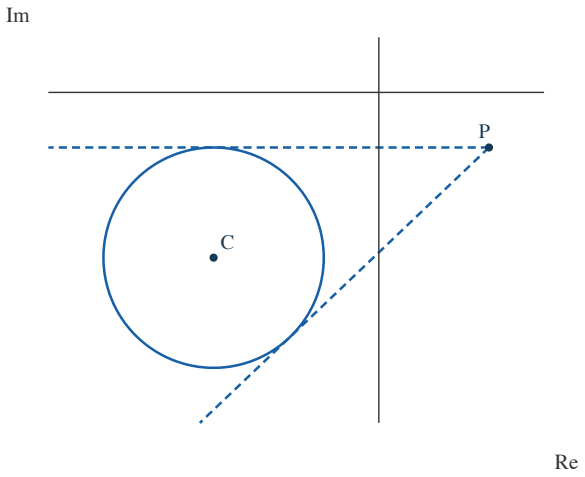
$\min|z|=6\sqrt{2}-4$

$\max|z|=6\sqrt{2}+4$

b

The rays tangent from $(4,-2)$ have angles $-\pi$ and $-\pi+2\tan^{-1}(2/5)$. Thus no intersection occurs when

$-\pi+2\tan^{-1}(2/5)<\theta<\pi$



Regions — Solution 36

Shade the region satisfying all three conditions:

$|z-4-2i|\le 2$

$|z-4|<|z-6|$

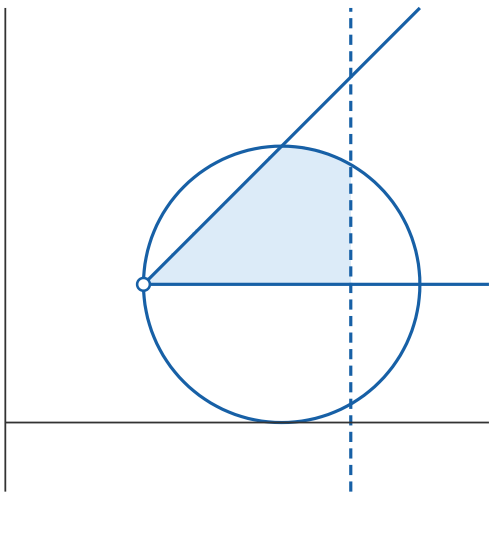
$0\le\arg(z-2-2i)\le\pi/4$

Region

Inside the circle centred at $(4,2)$, left of $x=5$, and between $y=2$ and $y=x$.

Exclude $x=5$ and the ray vertex $(2,2)$.

Sketch



Test Your Understanding — Solution 37

Shade

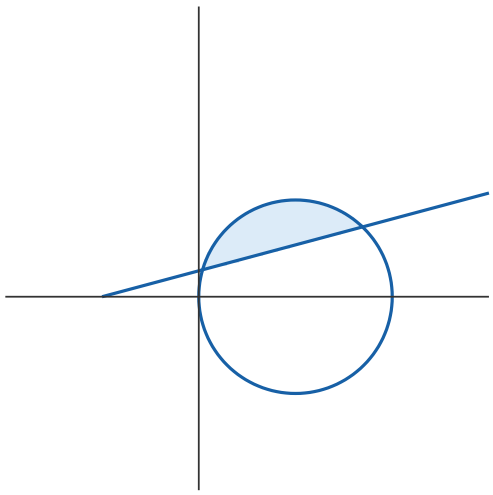
$|z-1| \leq 1, \quad \pi/12 \leq \arg(z+1) \leq \pi/2$

Region

Inside the circle centred at $(1,0)$, radius 1, and above the ray from $(-1,0)$ at angle $\pi/12$.

Sketch

Im



Re

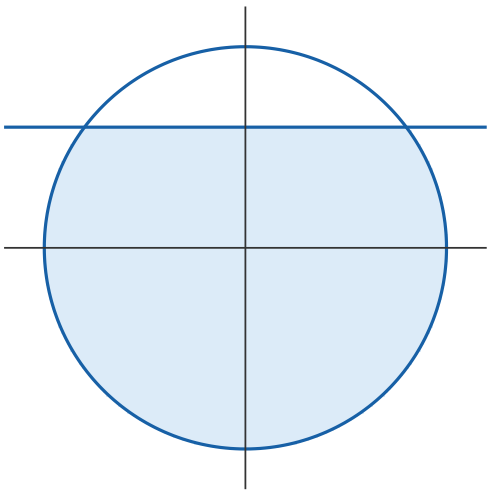
Exercise 2F Pearson Core Pure 1, p. 38 — Solution 38

2. Shade $|z| \leq 5$ and $|z| \leq |z-6i|$.

2 The disk of radius 5, restricted to $y \leq 3$.

Sketch

Im



Re

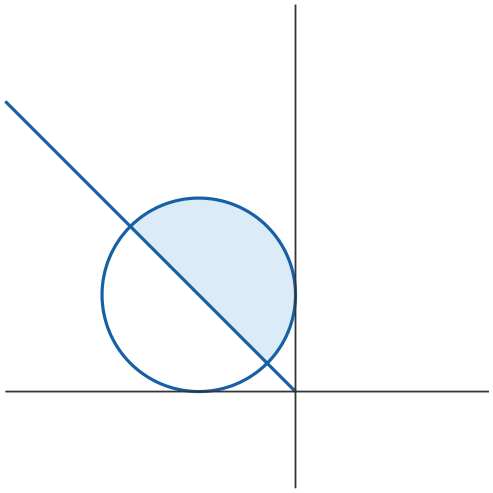
3. Shade $|z+1-i| \leq 1$ and $0 \leq \arg z \leq 3\pi/4$.

3

The disk centred at $(-1,1)$, restricted to $y \geq -x$.

Sketch

Im



Re

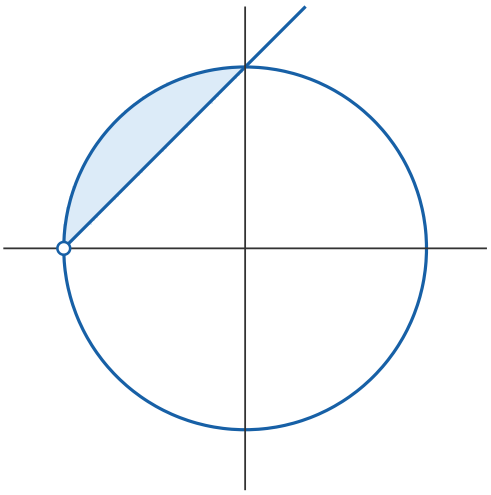
Exercise 2F Pearson Core Pure 1, p. 38 — Solution 39

4. Shade $|z| \leq 3$ and $\pi/4 \leq \arg(z+3) \leq \pi$.

Region

The part of the disk above $y = x + 3$. The point $(-3, 0)$ is excluded because the argument there is undefined.

Sketch



Exercise 2F Pearson Core Pure 1, p. 38 — Solution 40

5a. Sketch:

i. $|z-2| = |z-6-8i|$

ii. $\arg(z-4-2i) = 0$

iii. $\arg(z-4-2i) = \pi/2$

b. Shade $|z-2| \leq |z-6-8i|$ with $0 \leq \arg(z-4-2i) \leq \pi/2$.

a

i. $y = -x/2 + 6$

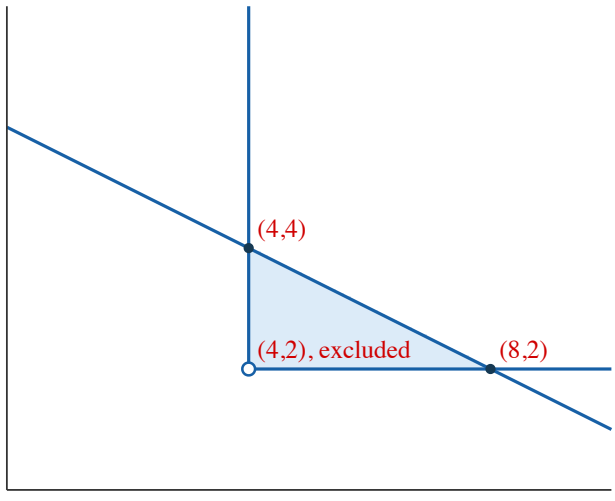
ii. $y = 2,\ x > 4$

iii. $x = 4,\ y > 2$

b

The triangle with vertices $(4, 2), (8, 2), (4, 4)$, including its sides except for $(4, 2)$.

Sketch



6a. Sketch $|z+10| = |z-6-4\sqrt{2}i|$ and $|z+1| = 3$.

b. Find their intersections.

c. Shade $|z+1| \leq 3$ with $|z+10| \leq |z-6-4\sqrt{2}i|$.

a

$y = -2\sqrt{2}(x+1)$

$(x+1)^2 + y^2 = 9$

b

$z = -2\sqrt{2}i$ or $z = -2+2\sqrt{2}i$

Sketch

